



Status Report

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Organization: KTH

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1. General description of status

Chinmay

- What has happened since the last report?
- What is the status?

2. Resource Status

2.1. Time and Effort

The project team has so far dedicated fewer hours than originally planned. This decision was made in response to delays from our supervisor (project sponsor) in providing the functional and synthesizable code base from last year's group, which was supposed to serve as the starting point for our work according to the project specifications.

Since the received code contained several compilation and synthesis errors, part of the allocated working hours was redirected to identifying and understanding these issues before further development. To avoid unnecessary effort and potential resource waste at this early stage, each team member limited their contribution hours during the first sprints.

The cumulative time contribution per member and sprint is summarized below:

Member	Sprint 0	Sprint 1	Sprint 2
Chongnan Wang	5h	8h	4h
Xinxin Qin	6h	8h	4h
Paul Martin	6h	9h	3h
Xiaotian Jiang	5h	8h	2h
Chinmay Purandare	7h	8h	3h
Lorenzo Parata	5h	7h	4h

Table 1: Time contribution per member and sprint.

In total, approximately **104 working hours** have been spent across all members during the first three sprints. In the upcoming sprints, we plan to significantly increase the number of hours dedicated to project development to compensate for the initial delay and ensure steady progress according to our agile workflow.

2.2. Budget and Economic Status

As this is an academic course project, no financial budget is involved. Team members' work is unpaid, and the necessary hardware resources are provided by the supervisor, likely obtained through educational sponsorships.

2.3. Personnel

The project currently consists of six active members, each contributing according to the agile sprint plan described above.

2.4. Materials and Equipment

Each team member is currently using one **DE0-Nano-SoC Development Kit**, a robust FPGA platform based on the Altera System-on-Chip (SoC) architecture. The board combines a dual-core ARM Cortex-A9 processor with programmable logic, enabling flexible and high-performance embedded system design. The kit includes DDR3 memory, analog-to-digital capabilities, and Ethernet connectivity, offering a complete environment for hardware-software co-design.

2.5. Deliverables

So far, the team has delivered an initial **testbench prototype**, developed during a one-week sprint by three members (Xiaotian, Xinxin, and Chongnan). Meanwhile, Chinmay, Lorenzo, and Paul have been focusing on identifying compilation and synthesis errors within the legacy code and exploring possible solutions to achieve a functional starting point for further design.

2.6. Outlook

In the next sprint sessions, we expect to increase the workload substantially, as the main development phase begins. The priority will be to stabilize the existing code, integrate a functional testbench, and progress toward the implementation and verification stages.

3. Problems / Action plan

- State the problems the sponsor and steering committee need to be aware of, and proposals for action

4. Risks / Action plan

On this part we will follow-up and update the risk analysis we previously did. It's important here to state, as much as possible, the risks the sponsor and steering committee need to be aware of, and to propose actions to reduce those ones.

4.1. Follow-up of Risk Analysis

The initial risk analysis was performed at the beginning of the project, however it is necessary to re-evaluate the risks we listed and possibly add new ones. That is why we try to follow up on risks and update the risk matrix when new issues appear.

The mitigation strategies defined in the initial risk analysis have already proven his use in practice. Indeed, several of the identified risks have already occurred but were mitigated.

For example:

- **Toolchain dependency** - we have tested the compiler early and it helped us detect errors quickly to manage them.
- **Hardware issues** - we also tested the hardware as soon as possible and we have seen non-functional cable and power adapter incompatibilities that we succeeded to solve without delaying development.
- **Environment setup** - for some of us with unusual systems or machines, this part wasn't that easy. And sharing documentation and tips within the group ensured that all members have a working FPGA and synthesis environment.

4.2. Updated Risk Analysis

We decided to follow up on the risk analysis and update it using the mini risk method.

ID	Risk	Description	P	C	R	Suggested Action	Follow-up Date	Trend
1	Hardware complexity	Two processor models on separate FPGAs increase design and debugging effort.	3	4	12	Start with single-FPGA setup, define early comm. protocols.	31 Oct	↓
2	Multi-FPGA system complexity	Communication and timing between boards difficult to synchronize.	4	4	16	Test modules separately, simulate before hardware link.	31 Oct	→
3	Extreme environment	SoI may not survive Venus-like temperatures.	2	5	10	Study environmental limits; keep simulation-level validation.	31 Oct	→
4	High development cost	ASIC fabrication and tooling are expensive.	1	3	3	Focus on FPGA prototyping only.	17 Oct	↓
5	Performance gap	Bit-serial version may not reach required speed.	4	3	12	Define realistic targets, compare to parallel baseline.	31 Oct	→
6	Verification coverage	Testbench may not provide full functional testing.	3	5	15	Add random/stress test patterns; document coverage.	31 Oct	→
7	Specialized toolchain dependency	ProGram compiler unstable or limited.	3	3	9	Validate early, keep VHDL rewrite backup plan.	24 Oct	↓
8	Novel microarchitecture flaws	CISC-V model may contain logical deadlocks.	4	2	8	Write detailed spec, co-simulate with golden model.	24 Oct	↓
9	Members' schedules differ	Difficult to meet as a full team.	5	2	10	Assign rotating secretaries, record meeting minutes.	17 Oct	↓
10	Internal conflicts	Potential disagreements between members.	3	4	12	Address through discussion or reassigning tasks.	31 Oct	→
11	Integration timeline under-estimation	System-level testing may take longer than expected.	5	3	15	Allocate buffer time, set MVP milestone.	31 Oct	→
12	Supervisor unavailability	Feedback delayed due to limited availability.	3	2	6	Plan meetings in advance, prepare written questions.	17 Oct	↓
13	Baseline design quality unknown	Legacy design may contain bugs.	4	2	8	Perform early code review and functional tests.	24 Oct	↓

Table 2: Mini Risk Method Table

Impact \ Likelihood	Rare	Unlikely	Possible	Likely	Almost Certain
Severe		3	6		
Major			1, 10 ↓	2	
Moderate	4 ↓		1, 7 ↓	5	11
Minor			7, 12 ↓	8, 13 ↓	9 ↓
Insignificant	4		12	8, 13	9

Figure 1: Updated Risk Matrix

4.3. Risks to Be Reported

According to the latest risk matrix evaluation, four risks are currently considered as really critical for the project (in the red zone) and should have a higher attention.

Risk 3 – Extreme Environment

The SoI CMOS technology used in this project may not be enough for the environmental conditions of Venus. If it's the case, the final demonstration may not be really representative of the objective and it would limit the scientific value of the prototype. We should keep in mind this while creating the testbench to imagine a way to simulate as closely as possible the Venus environmental conditions.

Risk 2 – Multi-FPGA System Complexity

The project requires two FPGAs, one for the processor and one for the testbench, communicating through a custom interface. We haven't done this communication interface yet but it might be the most challenging part of the project. Synchronization, timing management, and debugging of this inter-FPGA are difficult and could lead to delay in the final validation phase. That is why we have allocated several sprints for this task.

Risk 6 – Verification Coverage

The testbench might not generate a sufficiently diverse set of test scenarios to achieve full verification coverage. A lot of special cases should be taken care of and an incomplete verification may allow hidden functional bugs. Discovering those bugs in the late stages of the project could impact the demonstration reliability. As a result, we will try to expand the testbench with randomized and stress testing sequences.

Risk 11 – Integration Timeline Underestimation

Multi-FPGA integration may take longer than expected. This phase depends on several things: synthesis, communication debugging, and testbench validation. Those delays could compress the verification phase and result in a rushed testing, which is never good in a project. It's important to have a structured plan with clear milestones, define a "minimum viable demo" to ensure at least one stable version at the end and allocate additional buffer time in the coming sprints to focus on debugging and hardware validation.

5. Project changes from the project plan

- This is where changes are documented so that they become visual. Changes are typically: new requirements, changed requirements, changed organization, changed prerequisites, new documents, etc.

6. Appendix

- Updated time plan
- Updated resource plan